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THIN-FILM FLOW OVER A ROTATING PLATE: AN ASSESSMENT OF THE SUITABILITY OF VOF AND EULERIAN THIN-FILM METHODS FOR THE NUMERICAL SIMULATION OF ISOTHERMAL THIN-FILM FLOWS

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Abstract
An isothermal thin-film flow over a rotating plate has been simulated using the depth-averaged Eulerian Thin-Film modelling (ETFM) approach. The model setup is based on published experimental and numerical Volume of Fluid (VOF) CFD studies of the same problem to allow for model validation. A range of controlled film inlet heights and mass flow rates are explored together with varied plate rotational speeds ranging from a stationary plate (50rpm) to 200 rpm. While the VOF model has previously been shown to accurately reproduce film thickness, the Eulerian thin-film model is shown to provide predictions of comparable accuracy at a much lower computational cost. The model is also shown to be able to reproduce the film solution's sensitivity to variations in fluid properties due to changes in inlet temperature. A full 3D domain has been used in this study and the ETFM model is also shown to be able to reproduce azimuthal film thickness variations and surface features similar to those previously observed in experiments.

1. Introduction
Aeroengines and gas turbine oil systems involve multiphase flows of varying scale ranging from thin-film flows to much thicker flows within the same system. A thin-film may be described as a layer of liquid partially bounded by a solid substrate, with a free surface where the liquid is exposed to another fluid - usually a gas [1]. In many applications, the depth of the thin-film is typically orders of magnitude smaller than the other lateral dimensions of the liquid expanse. In addition, the flow within the film is two-dimensional with the vertical flow component being negligible compared to the lateral flow.

The numerical simulation of thin-film flows poses a particular engineering challenge due to the high computational cost that would be required to resolve the film when using more conventional multi-phase flow methods, such as the volume-of-fluid (VOF) approach [2], which are suitable for the other larger scale flows within the same system. Traditional multiphase flow methods such as VOF require the use of very fine computational grids (as illustrated in Figure 1) in order to adequately resolve the film velocity and temperature profiles as well as the film interface and its curvature. This often results in a prohibitively high computational cost for many industrial thin-film flows.

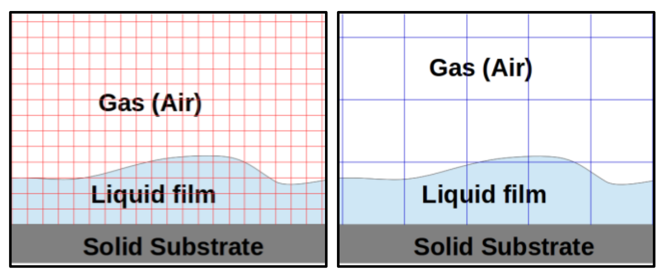

Figure 1: Illustration of film and interface resolution requirements for VOF (left) and ETFM (right) approaches to thin-film modelling

In order to address this constraint, the Eulerian Thin-Film modelling (ETFM) approach has been presented by a number of authors such as [3, 4, 5] as an alternative. A depth-averaging approach is often adopted, in which the three-dimensional (3D) Navier-Stokes equations are depth-averaged across the film thickness to obtain a set of 2D thin-film equations such as in [3, 4, 5] . These thin-film models allow for thin-film effects on the core flow and heat transfer to be accounted for without the need to explicitly resolve the very small scale film flows using a numerical grid. Although a number of numerical challenges remain with regards to shock and pool type film flow regimes in particular [6], these Eulerian Thin-Film Models (ETFM) present a unique opportunity for the reliable engineering simulation of thin-film flows at a reasonable computational cost.

There is however limited research comparing the performance of the two approaches – VOF and Eulerian Thin Film – for the simulation of thin-film flows. In this paper, the Eulerian thin-film model in commercial CFD code ANSYS FLUENT has been used to simulate the two-dimensional isothermal thin-film flow over a rotating plate. The resulting thin-film thickness predictions are contrasted against existing experimental data and VOF simulation results reported in [7]. The results are used to contrast the relative performance of both models and assess the suitability of the ETFM for such flows.

2. Problems setup

The computational domain used in this paper is taken from the experimental and VOF simulations study carried out by [7].

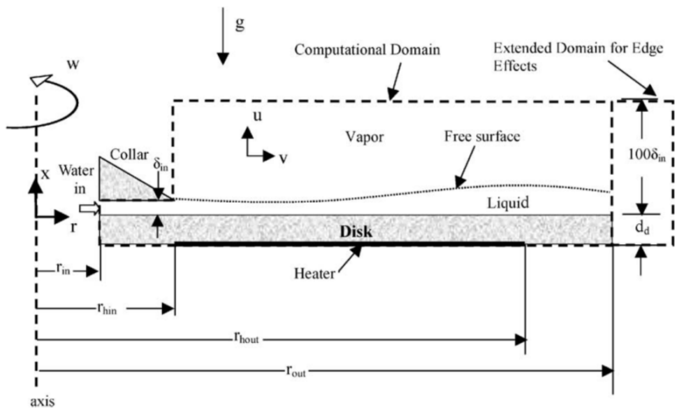

Figure 2: Schematic of experimental setup and computational domain used to model controlled liquid impongement on a disk by [7]. (Source: [7])

The domain consists of a horizontal rotating plate with an inner radius r_{in} = 0.0254 m and outer radius r_{out} = 0.2032 m. An initial film height δ_{in} = 0.254 mm has been used in all the cases. The plate is simulated as a full two-dimensional surface as opposed to a 1-D axisymmetric surface in [7]. A 2D unstructured quadrilateral mesh of 4200 faces is used to discretise the disk surface as illustrated in Figure 3.

The film fluid is assumed to be water at 40°C with a density and viscosity of 992.2 kg/m^3 and 6.53E-4 Pa.s respectively. In order to assess the sensitivity of results to inlet temperature a single is run with an inlet temperature of 20°C and corresponding fluid density and viscosity of 998.2 kg/m3 and 1.003E-3 Pa.s. A fixed mass flux is imposed at the inlet with a varied rate of either 7.0 lpm or 15.0 lpm.

The film depth is measured using a non-intrusive laser light interface reflection technique described in [8] where a typical error in the liquid film thickness measurements was estimated to be ±0.025 mm which is approximately ±10% of the initial collar film thickness of 0.254 mm.

A plate rotational speed is imposed at the plate wall boundaries. Rotational speeds of 50 rpm, 100 rpm and 200 rpm have been simulated.

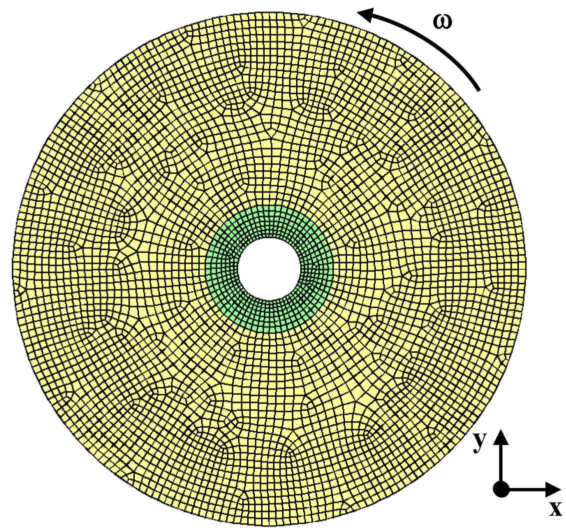

Figure 3: Unstructured hexahedral grid used to discretise the disc surface. Plate inlet region (beneath the collar) is shown in green.

The film is driven by a wall shear stress due to the plate rotation together with an interfacial shear stress due to the film-vapour interaction. The interfacial shear stress is estimated using a film-vapour phase coupling described in section 4. The film is also subjected to gravity, pressure gradient and surface tension forces. Gravity acts in the direction normal to the plate as illustrated in Figure 2.

The following section describes the thin-film model used in this study, which is after [7] and implemented in the commercial CFD code ANSYS FLUENT [8].

3. Thin-film modelling

The thin-film flow over a rotating plate is idealised as an incompressible Newtonian liquid of density, ρ_l and viscosity, μ_l flowing over a solid substrate and with a free-surface exposed to an incompressible Newtonian gas of ρ_g and viscosity, μ_g. The film has a spatially varying height, $h(x,y,t)$ and flows with a film velocity $\boldsymbol{u}(x,y,z,t)$ - where x and y are the lateral flow direction and z is the vertical direction normal to the surface. The film velocity $\boldsymbol{u}(x,y,z,t)$ may be decomposed into a depth-averaged component $\bar{\boldsymbol{u}}(x,y,t)$ and a fluctuating component $\hat{\boldsymbol{u}}(x,y,z,t)$ as:

$$\boldsymbol{u}(x,y,z,t) = \bar{\boldsymbol{u}}(x,y,t) + \hat{\boldsymbol{u}}(x,y,z,t), \qquad (1)$$

where the mean film speed, $\bar{\boldsymbol{u}}(x,y,t)$ components $(\bar{u}_x(x,y,t), \bar{u}_y(x,y,t))$ are computed according to

$$\bar{u}_i(x,y,t) = \frac{1}{h}\int_0^h u_i(x,y,t)dy. \qquad (2)$$

Subscript i denotes the x- and y-directions. A volumetric film flux may then be computed for each of the x and y flow directions as, $q_i = \bar{u}_i h$. This film flux, q, together with the film height define the film state which is the result of an interaction between viscous forces, gravity ($\boldsymbol{g}$ = (gx, g_y, g_z)), hydrostatic pressure, surface tension forces and film inertia. The resulting film flow dynamics over the solid-substrate may be described by the depth averaged continuity and momentum equations given by [4]:

$$\frac{\partial h}{\partial t} + \frac{\partial q_i}{\partial x_i} = 0, \tag{3}$$

$$\frac{\partial q_i}{\partial t} + \frac{\partial \bar{u}_i \bar{q}_i}{\partial x_i} = -\frac{h}{\rho_l}\frac{\partial P_l}{\partial x_i} + \frac{h}{\rho_l}\frac{\partial \sigma \kappa_i}{\partial x_i} + g_i h + S_{\tau i}\,, \tag{4}$$

where $P_l = (P_{\text{gas}} - \rho g_z h)$, is the film pressure which has a component from the interfacial vapour pressure, P_{gas}, and the film hydrostatic pressure, $\rho g_z h$. P_l is used to compute the film hydrostatic pressure gradient term, which is the first term on the right hand side (R.H.S.) of Equation 4.

Surface tension effects are represented in the surface tension term, $\left(\frac{h}{\rho_l}\frac{\partial \sigma \kappa_i}{\partial x_i}\right)$, where σ is the surface tension coefficient for the liquid-gas interface, and the interface normal curvature, κ_i, is estimated according to;

$$\kappa_i = \frac{\frac{d^2 h}{d{x_i}^2}}{\left(1 + \left(\frac{dh}{dx_i}\right)^2\right)^{1.5}} \approx \frac{d^2 h}{d{x_i}^2}. \tag{5}$$

The third term on the R.H.S. of Equation 4 represents the momentum source term due to film gravitational body forces in the direction of the film flow. Finally, the fourth source term on the R.H.S. of Equation 5, S_τ, represents the balance of viscous shear forces on the film, including contributions from the interfacial shear stress driving the film, $\boldsymbol{\tau_a} = (\tau_{ax}, \tau_{ay})$, and the wall shear stress resisting fluid flow over the solid substrate, $\boldsymbol{\tau_a} = (\tau_{wx}, \tau_{wy})$, Using appropriate sub-models to estimate $\boldsymbol{\tau_a}$ and $\boldsymbol{\tau_w}$, the viscous source term, $\boldsymbol{S_\tau}$ may be computed according to:

$$S_{\tau i} = \frac{\tau_{ai} - \tau_{wi}}{\rho_l} \tag{6}$$

In this paper, $\boldsymbol{\tau_a}$ has been computed from a direct coupling of the thin-film model with the model for the background air flow described in section4. The inertia/convective term of Equation 4 is representing using a simplified inertia approximation which for simplicity assumes that $\frac{\partial}{\partial s}\int_0^h u^2 dy = \frac{\partial h \bar{u}\bar{u}}{\partial s}$.

This “simplified inertia” approximation present in previous thin-film models such as [4] and also in the commercial code ANSYS FLUENT’s thin-film model [8] is however unrealistic under conditions of a strong driving shear at the film interface with a no-slip wall condition. The implications of this inertia simplification on the stability and accuracy of resulting thin-film solutions is explored in more detail in a separate paper [6].

An explicit time scheme is selected for discretising the unsteady. All convective terms are discretised using a second order upwind scheme while the diffusive terms such as the surface tension and pressure gradient term are discretised using a second order accurate central differencing scheme.

4. Single-phase air flow modelling

The velocity and pressure in the vapour region above the film interface are obtained by solving the 3D steady incompressible Navier Stokes equations (Equations 7 and 8) in commercial code ANSYS FLUENT 15.0.7 [7]:

$$\frac{\partial u_i}{\partial x_i} = 0\,, \tag{7}$$

$$\frac{\partial u_i u_j}{\partial x_j} = -\frac{1}{\rho}\frac{\partial P_{\text{gas}}}{\partial x_i} + \frac{\mu}{\rho}\frac{\partial^2 u_i}{\partial x_j \partial x_j} + f_i\,, \tag{8}$$

Where P_{gas} is the gas pressure, ρ is the vapour density and viscosity, μ is the viscosity. Air properties have been assumed for the vapour phase and no evaporation is considered in the current paper. The vapour domain is discretised using an unstructured hexahedral mesh of 30500 cells. The external boundaries of the vapour region have all be modelled as pressure outlets.

A realizable k-ε turbulence model [9] is used to model the core flow turbulence properties. The convective terms are discretised using a second order upwind scheme while the PRESTO scheme is used for pressure and the SIMPLE scheme for pressure-velocity coupling is used.

To account for the non-linear film-vapour phase interaction, the wall shear stress from the vapour phase model is used to drive the thin-film flow at the interface, $\boldsymbol{\tau_a}$. In addition, the wall velocity in the core flow simulation is assumed to correspond to the film surface velocity rather than the wall velocity, except during start-up where a dry wall is present.

The core flow has been simulated as a steady flow with the thin-film model simulated as an unsteady flow in order to ensure its numerical stability. The two models are coupled through the exchange of interfacial shear stress and velocity information at each core flow iteration. The simulation is then run until both the core flow and film fluid properties reach a steady state.

The following section discusses key results.

5. Results

A total of seven simulation cases have been run covering different combinations of plate rotational speed and film mass flux. These are summarised in Table 1. Case 4 was run to assess the sensitivity of results to film inlet temperature.

Table 1: Summary of wall-film simulation cases run

Case	Disk Speed [RPM]	Film flux [LPM]	Inlet Temp [°C]
1a	50	7.0	40.0
2a	100	7.0	40.0
3a	200	7.0	40.0
1b	50	15.0	40.0
2b	100	15.0	40.0
3b	200	15.0	40.0
4	50	15.0	20.0

Figure 4 shows the steady state film-thickness distribution on the plate for cases 1b, 2b and 3b of Table 1. The solutions predicted are characterised by smooth film profiles with gradual changes in film thickness. The overall maximum film thickness is shown to decrease with increasing plate rotational speed due to increased centrifugal force acting on the film.

The average solution time for these cases (Table 1), starting from a dry wall up to the film steady state was up to **15 minutes** on eight (8) 2.4GHz Intel Xeon E5620 cores, when using the ETFM. In comparison, a 2D axisymmetric VOF case run on the same numerical grid as specified in [7] and using ANSYS FLUENT on the same hardware platform, took **3 hours and 45 minutes** to reach a steady state solution starting from the same dry wall initial condition as in the ETFM cases. The Eulerian Thin-Film Model (ETFM) therefore represents a considerable gain in computational speed and reduction in mesh resolution requirements for thin-film flow simulation. Further improvements to this model to allow the accurate simulation of more complex shock and pooling film profiles is therefore merited.

With the ETFM approach, 3D film surface features observed in the experiments [8, 7] are resolved (Figure 4) which the 2D axisymmetric VOF approach does not capture. These features can be key in thin-film flows where surface tension terms dominate in certain regions of the flow. However, it is reasonable to expect that with a larger 3D film resolving VOF model capable of representing surface features, the computational cost would be even further increased compared to the ETFM approach.

In order to validate the thin-film results, the mean thin-film profile variation in the radial direction has been extracted and compared against existing experimental measurements and 2D axisymmetric VOF simulation results from [7]. This comparison is illustrated in Figure 5 for cases 1a – 3a and 1b – 3b. In all cases, the thin-film profile predictions compared well with the experimental measurements and VOF results. The thin-film model was able to accurately predict the peak film thicknesses, their radial location as well as the general film morphology changes due to changing rotational speed and mass flux. In some cases, such as Case 1b, the much more computationally efficient thin-film model gave a better agreement with the experimental results than the VOF simulations from [7].

A quantitative comparison between the performance of the ETFM and VOF predictions are summarised in Table 2 using the Normalised RMSE (NRMSE) in the film thickness to quantify the percentage deviation from the experimental data. The NRMSE is computed according to Equation 9.

$$\mathrm{NRMSE}(h) = \frac{1}{(\hat{h}_{\max} - \hat{h}_{\min})}\sqrt{\frac{\sum_{i=1}^{n}(h_i - \hat{h}_i)^2}{n}}, \qquad (9)$$

Where h is the predicted film thickness profile data and $\hat{h}$ is the corresponding experimental film thickness data from [7]. The results in Table 2 show that for all cases except case 2b, the ETFM gives better agreement with the experimental observation as indicated by the lower NRMSE, compared to the VOF predictions. In some cases, the error on the VOF solution was more than twice that on the ETFM solution. However relatively similar accuracy was obtained in case 2b where the VOF NRMSE was lower than the ETFM NRMSE.

Table 2: Normalised Root-Mean-Square-Error (NRMSE) of the ETFM and VOF predictions, from the equivalent experimental observations by [7]

Case	NRMSE - ETFM [%]	NRMSE - VOF [%]
1a	24.6	32.5
2a	29.5	33.8
3a	18.4	23.1
1b	23.6	52.4
2b	66.8	51.5
3b	31.5	33.7

In addition the sensitivity of results to inlet temperature has been assessed using case 4 and results are shown in Figure 6. Lower temperatures result in higher viscosity leading to a higher peak thickness and more film thickness variation.

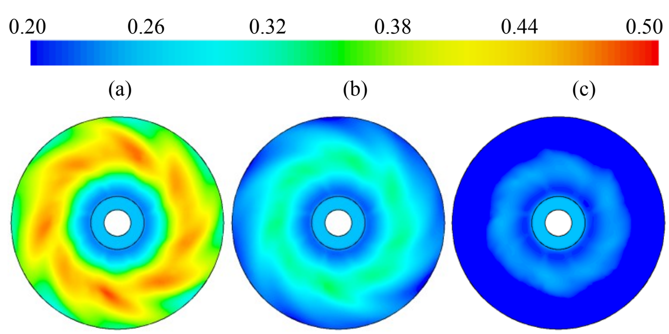

Figure 4: Steady state thin-film thickness profiles (in mm) for a 15LPM film flux and varying rotational speeds of; (a) 50 RPM in case 1b, (b) 100 RPM in case 2b, (c) 200 RPM in case 3b.

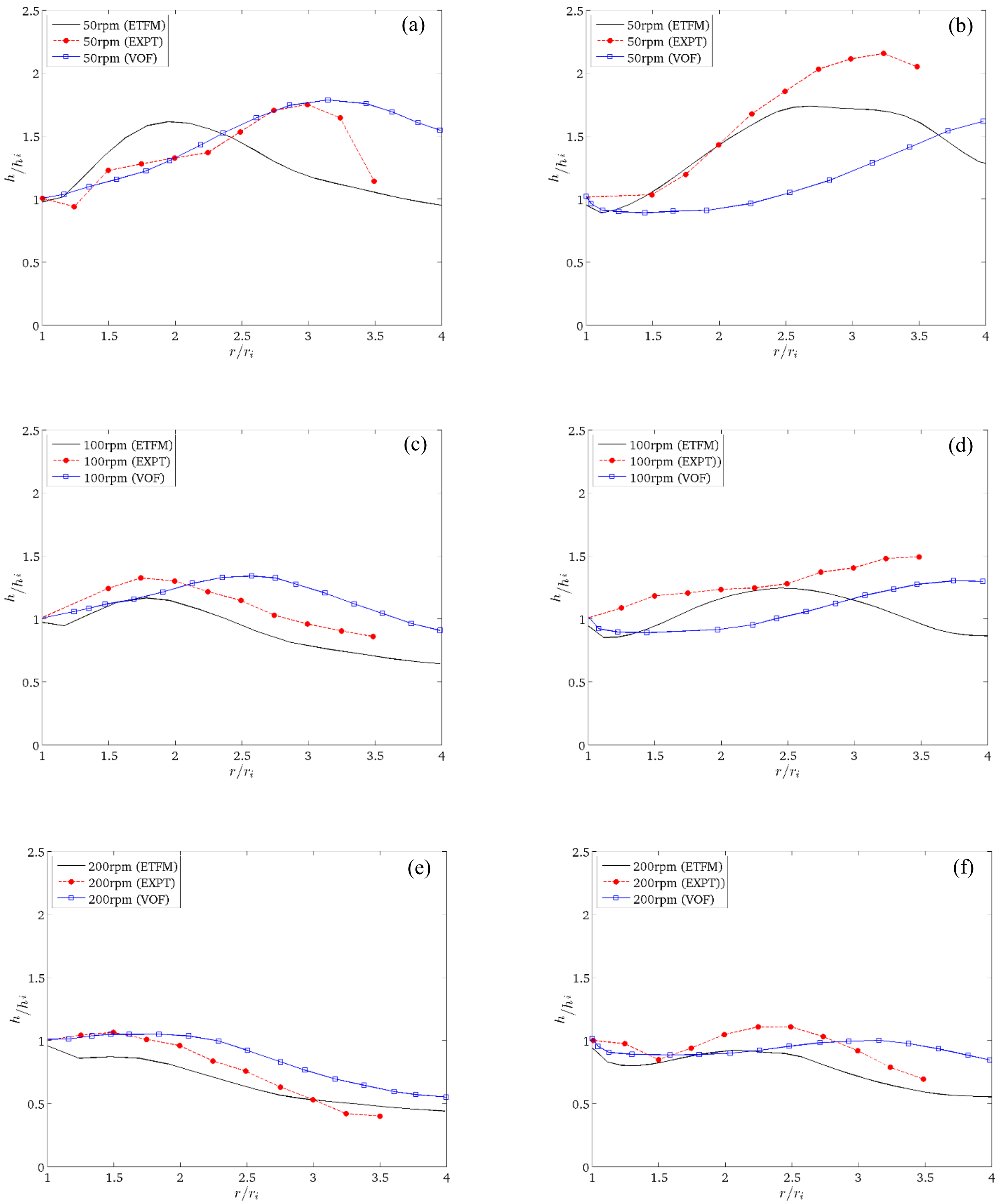

Figure 5: Normalised mean radial film thickness variation for; (a) case 1a (7LPM and 50RPM), (b) case 1b (15LPM and 50RPM), (c) case 2a (7LPM and 100RPM), (d) case 2b (15LPM and 100RPM) , (e) case 3a (7LPM and 200RPM), (f) case 3b (15LPM and 200RPM). Where h_i and r_i are the inlet film thickness and radius respectively, which are used to normalise the film profile.

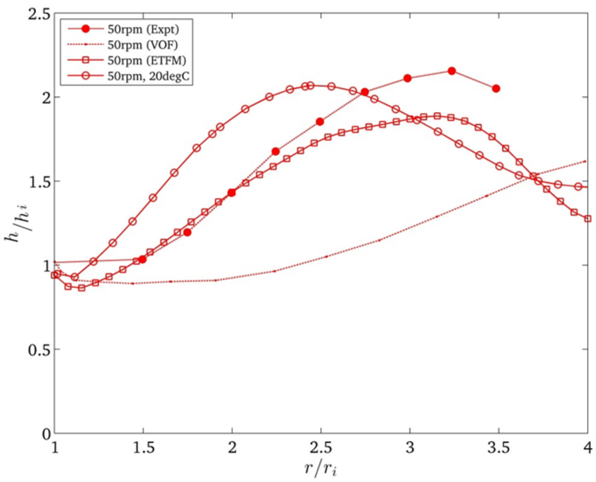

Figure 6: Thin-film flow prediction for cases 4 and 1b showing the influence of film inlet temperature on film profile

6. Conclusion

A Eulerian thin-film model (ETFM) has been successfully coupled with a steady state Finite Volume solution of the single phase vapour flow above a film in order to simulate the thin-film flow over a rotating plate.

For uni-directional flows with no vertical velocity component normal to the wall and smooth film profiles, the ETFM with a simplified inertia formulation offers a computationally efficient simulation approach. It allows for the use of coarser numerical meshes without the need to resolve film thickness with the grid leading to considerable computational savings.

Comparison of thin-film predictions against experimental data have shown that the ETFM model gives a comparable accuracy to the more computationally expensive 2D axisymmetric VOF CFD calculations.

The thin-film solution is shown to be sensitive to inlet temperature as previously presented in VOF studies such as [7]. The ETFM has been able to reproduce the effects of the thermal sensitivity of film fluid properties on film thickness over a rotating disk. This capability is particularly valuable for applications to thin-film flow and heat transfer in varying temperature environments.

Further ETFM investigations of this problem are recommended using more complete thin-film formulations (such as [3, 5] that incorporate a more complete inertia formulation in order to assess the impact of inertia simplification on solution accuracy. In addition, although the work presented here is primarily isothermal, it is recommended that non-isothermal effects be explored to assess the suitability of ETFM for predicting heat transfer.

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